

Solution 1

Solution 1 (a)

The Gaussian (Normal) distribution $N(\mu, \sigma^2)$ is symmetric about its mean μ .
Given $X \sim N(10, 16)$, it follows that the mean is $\mu = 10$.

$$P(X \geq \mu) = 0.5$$

Hence, the probability of the random variable X being greater than or equal to the mean is exactly 0.5, since the Gaussian distribution is symmetric about its mean.

$\therefore P(X \geq 10)$ is 0.5

Solution 1

Solution 1 (b)

The Gaussian distribution is a continuous probability distribution. For any continuous random variable, the probability of it taking a single, exact value is zero.

$\therefore P(X = 10)$ is 0

Solution 1

Solution 1 (c)

We can solve this using the Empirical Rule (68-95-99.7 rule). Given $X \sim N(10, 16)$, we have $\mu = 10$ and $\sigma = \sqrt{16} = 4$. We want to find $P(X \geq 14)$, which is $P(X \geq \mu + \sigma)$.

The Empirical Rule states that approximately 68% of the data lies within one standard deviation of the mean.

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68$$

$$P(6 \leq X \leq 14) \approx 0.68$$

The total probability outside this range is $1 - 0.68 = 0.32$.

$$P(X < 6) + P(X > 14) \approx 0.32$$

By the symmetry of the Gaussian distribution, the two tail probabilities are equal.

$$P(X \geq 14) = P(X > 14) \approx \frac{0.32}{2} = 0.16$$

$$\therefore P(X \geq 14) \approx 0.16$$

Solution 1

Solution 1 (d)

We use the Empirical Rule. Given $X \sim N(10, 16)$, we have $\mu = 10$ and $\sigma = 4$. We want to find $P(X \leq 2)$, which is $P(X \leq 10 - 8)$ or $P(X \leq \mu - 2\sigma)$.

The Empirical Rule states that approximately 95% of the data lies within two standard deviations of the mean.

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

$$P(2 \leq X \leq 18) \approx 0.95$$

The total probability outside this range is $1 - 0.95 = 0.05$.

$$P(X < 2) + P(X > 18) \approx 0.05$$

By symmetry, the two tail probabilities are equal. We are interested in the left tail.

$$P(X \leq 2) = P(X < 2) \approx \frac{0.05}{2} = 0.025$$

$$\therefore P(X \leq 2) \approx 0.025$$

Solution 2

Solution 2 (a)

$$\begin{aligned}\lambda &= \frac{\sum_{k=0}^{n \geq 9} k \cdot N_k}{\sum_{k=0}^{n \geq 9} N_k} \\&= \frac{22 \cdot 0 + 66 \cdot 1 + 106 \cdot 2 + 115 \cdot 3 + 85 \cdot 4 + 55 \cdot 5 + 28 \cdot 6 + 13 \cdot 7 + 10 \cdot 8 + 9 \cdot 0 + \dots}{500} \\&= \frac{0 + 66 + 212 + 345 + 340 + 275 + 168 + 91 + 80 + 0}{500} \\&= \frac{1577}{500} \\&= 3.154\end{aligned}$$

$\therefore$ the value of λ you choose is 3.154

Solution 2

Solution 2 (b)

Let $\lambda = 3.154$ as found in *part (a)*.

Poisson pmf Code

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import poisson
4
5 # Parameter from part (a)
6 lambda_val = 3.154
7
8 # k values from 0 to 10
9 k_values = np.arange(0, 11)
10
11 # Calculate Poisson PMF probabilities for each k
12 probabilities = poisson.pmf(k_values, lambda_val)
13
14 # Create the bar plot
15 plt.figure(figsize=(10, 6))
16 plt.bar(k_values, probabilities, color='skyblue', edgecolor='black')
17
18 # Add labels and title
19 plt.xlabel('k (Number of Calls in one hour)', fontsize=12)
20 plt.ylabel('Probability P(X=k)', fontsize=12)
21 plt.title(f'Poisson PMF for  $\lambda = \{lambda\_val\}$ ', fontsize=14)
22 plt.xticks(k_values) # Ensure integer ticks on x-axis
23 plt.grid(axis='y', linestyle='--', alpha=0.7)
24
25 # Save the figure
26 plt.savefig('poisson_pmf.png', dpi=150, bbox_inches='tight')
27
28 print("Poisson PMF plot saved as 'poisson_pmf.png'")
29 # Print probabilities for reference
30 for k, prob in zip(k_values, probabilities):
31     print(f"P(X={k}) = {prob:.5f}")
32
33 """
34 # output from for loop
35 P(X=0) = 0.04268
36 P(X=1) = 0.13462
37 P(X=2) = 0.21229
38 P(X=3) = 0.22319
39 P(X=4) = 0.17598
40 P(X=5) = 0.11101
41 P(X=6) = 0.05835
42 P(X=7) = 0.02629
43 P(X=8) = 0.01037
44 P(X=9) = 0.00363
45 P(X=10) = 0.00115
46 """
```

Solution 2

Solution 2 (b)

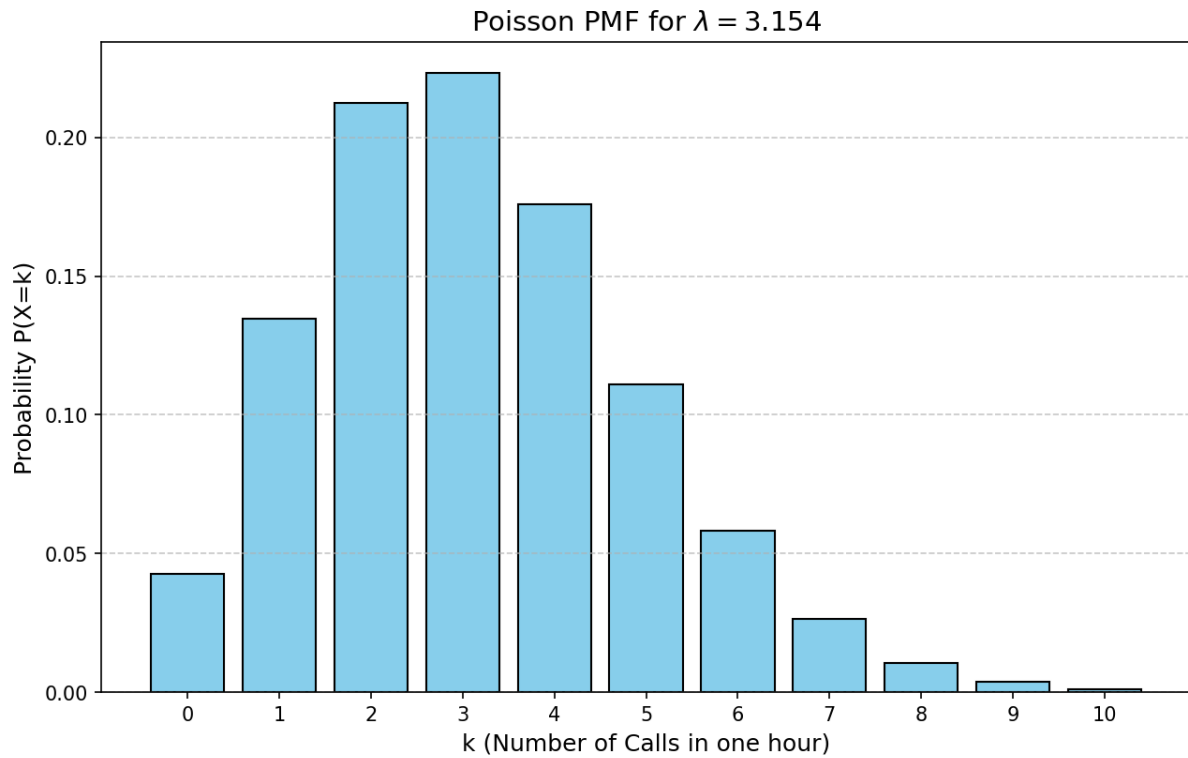


Figure 1: Poisson PMF distribution for $\lambda = 3.154$, showing $P(X = k)$ for $k = 0 \dots 10$.

Solution 2

Solution 2 (c)

Use the probabilities $P(X = k)$ from part (b) and multiply by 500 to get expected one-hour intervals with k calls.

$$\begin{aligned}
 P(X = 0) &= 0.04268 & \implies E[N_0] &= 500 \times 0.04268 = 21.34 \\
 P(X = 1) &= 0.13462 & \implies E[N_1] &= 500 \times 0.13462 = 67.31 \\
 P(X = 2) &= 0.21229 & \implies E[N_2] &= 500 \times 0.21229 = 106.15 \\
 P(X = 3) &= 0.22319 & \implies E[N_3] &= 500 \times 0.22319 = 111.60 \\
 P(X = 4) &= 0.17598 & \implies E[N_4] &= 500 \times 0.17598 = 87.99 \\
 P(X = 5) &= 0.11101 & \implies E[N_5] &= 500 \times 0.11101 = 55.50 \\
 P(X = 6) &= 0.05835 & \implies E[N_6] &= 500 \times 0.05835 = 29.18 \\
 P(X = 7) &= 0.02629 & \implies E[N_7] &= 500 \times 0.02629 = 13.15 \\
 P(X = 8) &= 0.01037 & \implies E[N_8] &= 500 \times 0.01037 = 5.19 \\
 P(X \geq 9) &= 1 - \sum_{k=0}^8 P(X = k) = 1 - 0.99478 = 0.00522 & \implies E[N_{\geq 9}] &= 500 \times 0.00522 = 2.61
 \end{aligned}$$

Expected counts table

Table 1: Poisson($\lambda=3.154$) probabilities and expected counts for 500 intervals

k	$P(X = k)$	$E[N_k]$
0	0.04268	21.34
1	0.13462	67.31
2	0.21229	106.15
3	0.22319	111.60
4	0.17598	87.99
5	0.11101	55.50
6	0.05835	29.18
7	0.02629	13.15
8	0.01037	5.19
≥ 9	0.00522	2.61

Solution 3

Solution 3 (a)

For any probability density function (PDF), the total area under the curve must integrate to 1.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

We are given the bounds of the probability density function $[0, \lambda]$, so the equation becomes:

$$\begin{aligned}\int_0^{\lambda} c dx &= 1 \\ c \cdot [x]_0^{\lambda} &= 1 \\ c \cdot (\lambda - 0) &= 1 \\ c \cdot \lambda &= 1 \\ c &= \frac{1}{\lambda}\end{aligned}$$

$\therefore$ the density of U_{λ} is $\frac{1}{\lambda}$ and we can express it as $f(x | \lambda)$:

$$f(x | \lambda) = \begin{cases} \frac{1}{\lambda} & \text{if } 0 \leq x \leq \lambda \\ 0 & \text{otherwise} \end{cases}$$

Solution 3

Solution 3 (b)

From *part (a)* U_λ on $[0, \lambda]$, the density is

$$f(x | \lambda) = \begin{cases} \frac{1}{\lambda} & \text{if } 0 \leq x \leq \lambda \\ 0 & \text{otherwise} \end{cases}$$

The likelihood function $L(\lambda)$ for a set of n independent observations $X = \{x_1, \dots, x_n\}$ is the product of the individual densities:

$$L(\lambda | X) = \prod_{i=1}^n f(x_i | \lambda)$$

For this likelihood to be non-zero, every single term $f(x_i | \lambda)$ must be non-zero. Based on the PDF definition, this means we must have $0 \leq x_i \leq \lambda$ for all $i = 1, \dots, n$.

If we choose a λ that is smaller than even one observation, say x_j , then $f(x_j | \lambda) = 0$, and the entire likelihood $L(\lambda) = 0$. This is the minimum possible likelihood, not the maximum.

It follows that, to have a non-zero likelihood, the choice of λ must be greater than or equal to all observations or:

$$\lambda \geq \max(x_1, \dots, x_n)$$

$\therefore$ the max likelihood choice of λ is $\max(x_1, x_2, \dots, x_n)$

Solution 4

Solution 4 (a)

Chi Squared Code

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import chi2
4
5 # degrees of freedom
6 k = 10
7
8 # range of x values
9 x = np.linspace(0, 30, 400)
10
11 # chi squared PDF values
12 y_pdf = chi2.pdf(x, df=k)
13
14 # make the plot and save
15 plt.figure(figsize=(10, 6))
16 plt.plot(x, y_pdf, 'r-', lw=2, label=f'Chi-squared PDF (k={k})')
17 plt.xlabel('X', fontsize=12)
18 plt.ylabel('Probability Density', fontsize=12)
19 plt.title(f'Chi-squared Distribution PDF (k={k} degrees of freedom)', fontsize=14)
20 plt.legend()
21 plt.grid(True, linestyle='--', alpha=0.7)
22 plt.savefig('chi_squared_pdf.png', dpi=150, bbox_inches='tight')
```

Plot

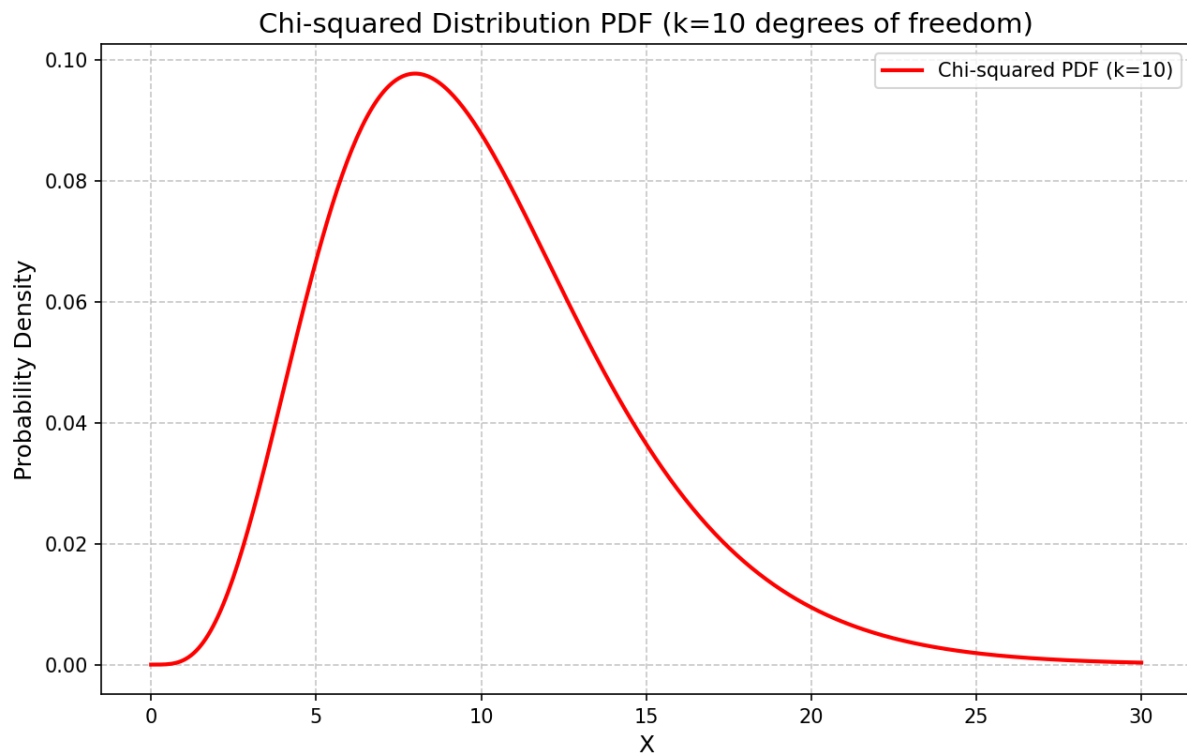


Figure 2: Chi-squared probability density function (PDF) with $k = 10$ degrees of freedom.

Solution 4

Solution 4 (b)

Chi Squared Median Code

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # 10000 samples from chi-squared(10) -> degrees of freedom is 10
5 k = 10
6 num_samples = 10000
7 random_seed = 42 # For reproducibility
8
9 # set the random seed
10 np.random.seed(random_seed)
11
12 # Generate 10,000 samples from chi-squared(10)
13 samples = np.random.chisquare(df=k, size=num_samples)
14
15 # Estimate the median by finding the median of the samples
16 median_estimate = np.median(samples)
17
18 print(f"Degrees of freedom (k): {k}")
19 print(f"Number of samples: {num_samples}")
20 print(f"Estimated Median: {median_estimate:.4f}")
21
22 # make plot
23 plt.figure(figsize=(10, 6))
24 hist_label = f'Histogram of {num_samples} samples'
25 plt.hist(samples, bins=50, density=True, alpha=0.7, label=hist_label)
26
27 # add vertical line at the estimated median to plot
28 median_label = f'Estimated Median: {median_estimate:.4f}'
29 plt.axvline(median_estimate, color='red', linestyle='--', linewidth=2,
30             label=median_label)
31
32 # set args for plot and save
33 plt.xlabel('X', fontsize=12)
34 plt.ylabel('Density', fontsize=12)
35 plt.title(f'Median Estimation from  $\chi^2(k={k})$  Samples', fontsize=14)
36 plt.legend()
37 plt.grid(True, linestyle='--', alpha=0.6)
38 plt.savefig('estimate_median.png', dpi=150, bbox_inches='tight')
```

Solution 4

Solution 4 (b)

Plot

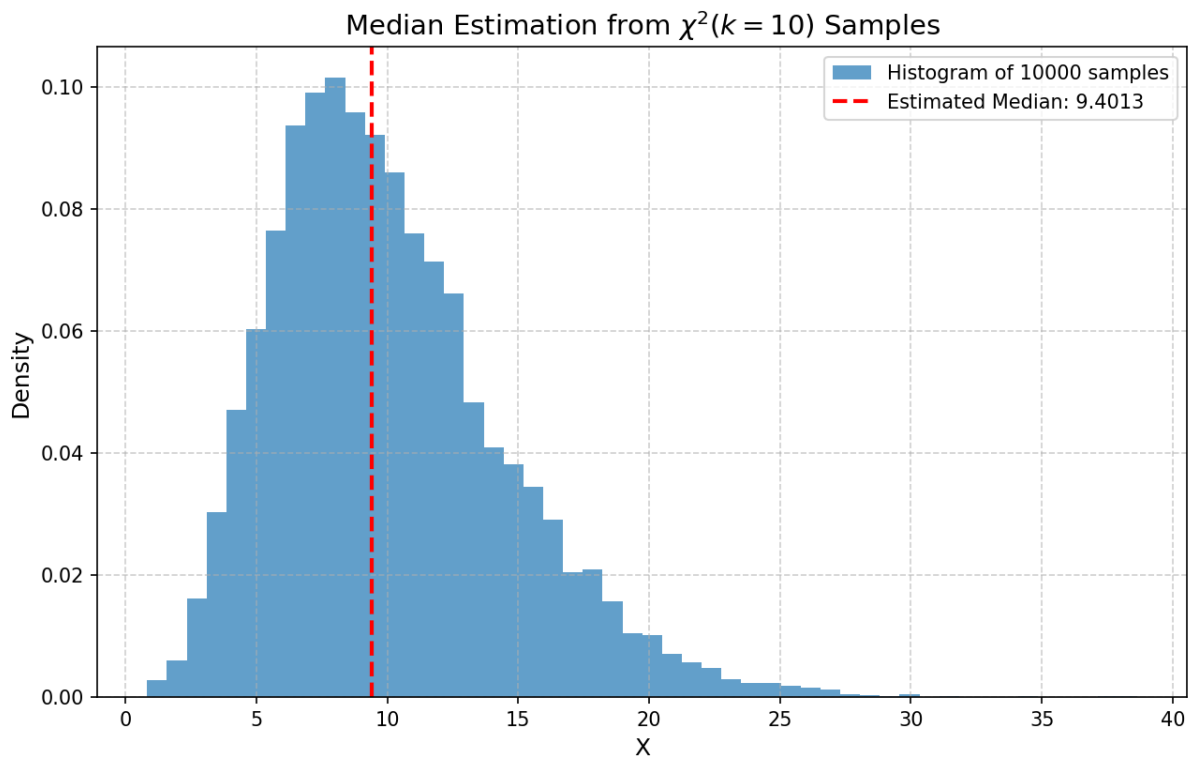


Figure 3: Histogram of 10,000 samples from a $\chi^2(10)$ distribution. The red dashed line shows the estimated median of the samples.

$\therefore$ The estimated median from 10,000 samples (with seed 42) is 9.4013.

Solution 5

Solution 5 (a)

The Maximum Likelihood (ML) estimate for the bias θ (probability of heads) of a coin is:

$$\theta_{\text{ML}} = \frac{n_H}{n} = \frac{n_H}{n_H + n_T}$$

Let,

$$n = 20 \quad n_H = 20 \quad n_T = 0$$

Substituting the values into the formula:

$$\begin{aligned}\theta_{\text{ML}} &= \frac{n_H}{n_H + n_T} \\ &= \frac{20}{20} \\ &= 1\end{aligned}$$

$\therefore$ the bias of the coin using maximum likelihood is: 1

Solution 5

Solution 5 (b)

For a coin (which has $k = 2$ outcomes: heads, tails), we add 1 to the count of heads and 1 to the count of tails to estimate the bias of a coin using the maximum likelihood with Laplace smoothing.

The formula becomes:

$$\theta_{\text{Laplace}} = \frac{n_H + 1}{n_H + n_T + 2}$$

Let,

$$n = 20 \quad n_H = 20 \quad n_T = 0$$

Substituting the values into the formula:

$$\begin{aligned}\theta_{\text{Laplace}} &= \frac{n_H + 1}{n_H + n_T + 2} \\ &= \frac{20 + 1}{20 + 0 + 2} \\ &= \frac{21}{22}\end{aligned}$$

$\therefore$ the bias of the coin using maximum likelihood with Laplace smoothing is: $\frac{21}{22}$

Solution 5

Question 5 (c)

From *part (b)*, our estimated bias (probability of heads) is:

$$\theta = P(H) = \frac{21}{22}$$

It follows that, the probability of tails is:

$$P(T) = 1 - \theta = 1 - \frac{21}{22} = \frac{1}{22}$$

The sequence of tosses is $HHTTTHH$. Now, calculate the probability:

$$\begin{aligned} P(HHTTTHH) &= P(H) \cdot P(H) \cdot P(T) \cdot P(T) \cdot P(H) \cdot P(H) \\ &= P(H)^4 \cdot P(T)^2 \\ &= (\theta)^4 \cdot (1 - \theta)^2 \\ &= \left(\frac{21}{22}\right)^4 \left(\frac{1}{22}\right)^2 \end{aligned}$$

$\therefore$ the probability of seeing the sequence of tosses $HHTTTHH$ is: $\left(\frac{21}{22}\right)^4 \left(\frac{1}{22}\right)^2$

Solution 6

Solution 6 (a)

Let,

$$D = \{a, rose, is, a, rose, is, a, rose\} \quad V = \{a, rose, is, flower\}$$

Now, count the occurrences of each vocabulary word in D :

$$\begin{aligned} |D_a| &= 3 \\ |D_{rose}| &= 3 \\ |D_{is}| &= 2 \\ |D_{flower}| &= 0 \end{aligned}$$

The vector form is an ordered vector of these counts, corresponding to the order of the vocabulary V .

$\therefore$ The vector form is $\begin{bmatrix} 3 \\ 3 \\ 2 \\ 0 \end{bmatrix}$, corresponding to the counts for (a, rose, is, flower).

Solution 6

Solution 6 (b)

The Maximum Likelihood (ML) estimate for the probability of a word w , θ_w , is the count of that word n_w divided by the total number of words in the document, N .

$$\theta_w = P_{\text{ML}}(w) = \frac{n_w}{N}$$

From *part (a)*, our counts are:

$$\begin{aligned} |D_a| &= n_a = 3 \\ |D_{\text{rose}}| &= n_{\text{rose}} = 3 \\ |D_{\text{is}}| &= n_{\text{is}} = 2 \\ |D_{\text{flower}}| &= n_{\text{flower}} = 0 \end{aligned}$$

The total number of words N is:

$$N = 3 + 3 + 2 + 0 = 8$$

We can now calculate the ML probabilities:

$$\begin{aligned} P_{\text{ML}}(a) &= \frac{n_a}{N} = \frac{3}{8} \\ P_{\text{ML}}(\text{rose}) &= \frac{n_{\text{rose}}}{N} = \frac{3}{8} \\ P_{\text{ML}}(\text{is}) &= \frac{n_{\text{is}}}{N} = \frac{2}{8} = \frac{1}{4} \\ P_{\text{ML}}(\text{flower}) &= \frac{n_{\text{flower}}}{N} = \frac{0}{8} = 0 \end{aligned}$$

$\therefore$ The resulting distribution is: $P(a) = \frac{3}{8}$, $P(\text{rose}) = \frac{3}{8}$, $P(\text{is}) = \frac{1}{4}$, $P(\text{flower}) = 0$

Solution 6

Solution 6 (c)

The formula for the smoothed probability of a word w is:

$$\theta_w = P_{\text{Laplace}}(w) = \frac{n_w + \alpha}{N + k\alpha}$$

With $\alpha = 1$, this simplifies to:

$$P_{\text{Laplace}}(w) = \frac{n_w + 1}{N + k}$$

Let,

$$n_a = 3 \quad n_{\text{rose}} = 3 \quad n_{\text{is}} = 2 \quad n_{\text{flower}} = 0 \quad N = 8 \quad k = |V| = 4$$

Now calculate the smoothed probabilities:

$$\begin{aligned} P_{\text{Laplace}}(a) &= \frac{n_a + 1}{N + k} = \frac{3 + 1}{12} = \frac{4}{12} = \frac{1}{3} \\ P_{\text{Laplace}}(\text{rose}) &= \frac{n_{\text{rose}} + 1}{N + k} = \frac{3 + 1}{12} = \frac{4}{12} = \frac{1}{3} \\ P_{\text{Laplace}}(\text{is}) &= \frac{n_{\text{is}} + 1}{N + k} = \frac{2 + 1}{12} = \frac{3}{12} = \frac{1}{4} \\ P_{\text{Laplace}}(\text{flower}) &= \frac{n_{\text{flower}} + 1}{N + k} = \frac{0 + 1}{12} = \frac{1}{12} \end{aligned}$$

$\therefore$ The resulting distribution is: $P(a) = \frac{1}{3}$, $P(\text{rose}) = \frac{1}{3}$, $P(\text{is}) = \frac{1}{4}$, $P(\text{flower}) = \frac{1}{12}$

Solution 7

Solution 7 (a)

Code

```

1  ## SOLUTION:
2  print("training bigram...")
3  bigram_model = create_and_fit_model(n=2,sentences=sentences)
4  print(f"print_generated_text(bigram): \n {print_generated_text(bigram_model)}")
5  print("training trigram...")
6  trigram_model = create_and_fit_model(n=3,sentences=sentences)
7  print(f"print_generated_text(bigram): \n {print_generated_text(trigram_model)}")
8
9  #===== OUTPUT =====#
10 training bigram...
11 =====
12 Generated text:
13
14
15 CLEOPATRA . De sin , to detain , so . My dear , Be patient ? BOY . Here she to fight .
    Thou shalt have been more above me ere Rome commanding Warwick rages , that love ,
    would it see . LUCENTIO with suitors , seeking that be now ? But to infringe our
    intent to go , by the while you , he inform 'd shall her modern friends ! GUIDERIUS
    . ARMADO . I should die is a prey of Saxony 's hare -finder and he would be good
    office from his looks . FORD and
16 =====
17 print_generated_text(bigram):
18     None
19 training trigram...
20 =====
21 Generated text:
22
23
24 Well , what , ho ! They vex me past care . I will watch to -night , Which on thy pale
    cold lips . Something sure of this child , Why then tomorrow night , waking and in
    such a face of her soilure , With ourselves . At my nativity to this princely
    presence To doom th ' hill . O Jupiter , I 'll give her patent to offend , either
    with ay or no . No blame belongs to a bowl of wine , and set Quarrelling upon the
    father 's moods . 'Tis dangerous to
25 =====
26 print_generated_text(bigram):
27     None
28 #===== END OUTPUT =====#

```

Solution 7

Solution 7 (b)

Python Implementation and Output

```
1 def model_ll(model, text):
2     """
3     Takes in an n-gram model and a sample text.
4     Returns the log-likelihood of the text under the given model.
5     """
6     import math
7     ll = 0
8     # raise NotImplementedError
9     ## SOLUTION:
10    # 1. Tokenize the text.
11    tokens = model.tokenize(text)
12
13    # 2. Generate ngrams for the tokens.
14    ngrams = model.generate_n_grams(tokens)
15
16    # 3. Loop through the ngrams, calculate log prob for each ngram and update ll.
17    for n_gram in ngrams:
18        ll += math.log(model.get_prob(context=n_gram[0], target=n_gram[1]))
19
20    return ll
21
22 text = "And on a love-book pray for my success?"
23 ## SOLUTION:
24 # train and fit unigram_model
25 unigram_model = create_and_fit_model(n=1, sentences=sentences)
26 ll = model_ll(unigram_model, text)
27 print(f"unigram log-likelihood: {ll:.3f}")
28 ll = model_ll(bigram_model, text)
29 print(f"bigram log-likelihood: {ll:.3f}")
30
31 #===== OUTPUT =====#
32 unigram log-likelihood: -65.861
33 bigram log-likelihood: -45.879
34 #===== END OUTPUT =====#
```

Solution 7

Solution 7 (c)

Code Output

```

1  # QUESTION
2  print(test_play)
3  prompt = "From fairest creatures we desire increase"
4  bigram_20 = generate_text(model=bigram_model, n_outs=20, prompt=prompt)
5  print(f"bigram 20 generated words: {bigram_20}")
6  trigram_20 = generate_text(model=trigram_model, n_outs=20, prompt=prompt)
7  print(f"trigram 20 generated words: {trigram_20}")
8
9  #===== OUTPUT =====#
10 THE SONNETS
11
12 by William Shakespeare
13
14
15
16             1
17 From fairest creatures we desire increase,
18 That thereby beauty's rose might never die,
19 But as the ripper should by time decease,
20 His tender heir might bear his memory:
21 But thou contracted to thine own bright eyes,
22 Feed'st thy light's flame with self-substantial fuel,
23 Making a famine where abundance lies,
24 Thy self thy foe, to thy sweet self too cruel:
25 Thou that art now the world's fresh ornament,
26 And only herald to the gaudy spring,
27 Within thine own buduriest thy content,
28 And tender churl mak'st waste in niggarding:
29     Pity the world, or else this glutton be,
30     To eat the world's due, by the grave and thee.
31 bigram 20 generated words: . I take away for the PRINCE OF EPHEsus . POLIXENES . How
    was my mistress , And y 'are
32 trigram 20 generated words: reconciled commenced Lo! burneth heraldry. augmentation
    'England Rais'd Mirth intemperance lo, persuasion. Man-ent'red his) French woman?
    dishonours black-nay, begot. LODOVICO,
33 #===== END OUTPUT =====#

```

Solution 8

Solution 8 (a)

The Maximum Likelihood Estimate (MLE) for a Bernoulli or Binomial process is:

$$\theta_{\text{ML}} = \frac{n_W}{n_W + n_L}$$

Let,

$$n_W = 5 \quad n_L = 5$$

Substituting into the equation

$$\begin{aligned}\theta_{\text{ML}} &= \frac{n_W}{n_W + n_L} \\ &= \frac{5}{5 + 5} \\ &= \frac{5}{10} \\ &= 0.5\end{aligned}$$

$\therefore$ maximum likelihood value of θ is 0.5

Solution 8

Solution 8 (b)

Let, the prior distribution $\theta \sim \text{Beta}(\alpha, \beta)$, with $\alpha = 6$ and $\beta = 4$.

Mean

The mean of a Beta distribution is given by the formula:

$$\mathbb{E}[\theta] = \frac{\alpha}{\alpha + \beta}$$

Substituting our values:

$$\mathbb{E}[\theta] = \frac{\alpha}{\alpha + \beta} = \frac{6}{6 + 4} = \frac{6}{10} = \frac{3}{5}$$

Standard Deviation

The variance of a Beta distribution is given by the formula:

$$\text{Var}(\theta) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

Substituting our values:

$$\text{Var}(\theta) = \frac{6 \times 4}{(6 + 4)^2(6 + 4 + 1)} = \frac{24}{(10)^2(11)} = \frac{24}{100 \times 11} = \frac{24}{1100}$$

The standard deviation is the square root of the variance:

$$\text{std}(\theta) = \sqrt{\text{Var}(\theta)} = \sqrt{\frac{24}{1100}}$$

$\therefore$ The prior mean is $\mathbb{E}[\theta] = \frac{3}{5}$ and the prior standard deviation is $\text{std}(\theta) = \sqrt{\frac{24}{1100}}$

Solution 8

Solution 8 (c)

Let the prior be $\theta \sim \text{Beta}(\alpha, \beta)$ with $\alpha = 6$ and $\beta = 4$, and suppose we observe $n_W = 5$ wins and $n_L = 5$ losses.

Posterior

Using Beta–Binomial conjugacy,

$$\begin{aligned}\alpha' &= \alpha + n_W = 6 + 5 = 11, \\ \beta' &= \beta + n_L = 4 + 5 = 9,\end{aligned}$$

so the posterior is $\theta \mid X \sim \text{Beta}(11, 9)$.

Mean

The mean of a Beta distribution is

$$\mathbb{E}[\theta \mid X] = \frac{\alpha'}{\alpha' + \beta'}.$$

Substituting our posterior parameters:

$$\mathbb{E}[\theta \mid X] = \frac{11}{11 + 9} = \frac{11}{20}.$$

Standard Deviation

The variance of a Beta distribution is

$$\text{Var}(\theta \mid X) = \frac{\alpha' \beta'}{(\alpha' + \beta')^2 (\alpha' + \beta' + 1)}.$$

Substituting our posterior parameters:

$$\text{Var}(\theta \mid X) = \frac{11 \times 9}{(11 + 9)^2 (11 + 9 + 1)} = \frac{99}{20^2 \cdot 21} = \frac{99}{8400}.$$

The standard deviation is the square root of the variance:

$$\text{std}(\theta \mid X) = \sqrt{\frac{99}{8400}}.$$

$\therefore$ The posterior distribution is $\theta \sim \text{Beta}(11, 9)$, the posterior mean is $\frac{11}{20}$, and the posterior standard deviation is $\sqrt{\frac{99}{8400}}$.

Solution 8

Solution 8 (d)

Code: density plot posterior distribution

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3  from scipy.stats import beta
4
5  # set parameters for the prior
6  alpha_prior = 6
7  beta_prior = 4
8
9  # set wins and losses
10 n_W = 5
11 n_L = 5
12
13 # set parameters for the posterior
14 alpha_post = alpha_prior + n_W
15 beta_post = beta_prior + n_L
16
17 # create np.array of x values (theta)
18 x = np.linspace(0, 1, 400)
19
20 # calculate the PDF for prior and posterior
21 prior_pdf = beta.pdf(x, alpha_prior, beta_prior)
22 post_pdf = beta.pdf(x, alpha_post, beta_post)
23
24 # calculate the likelihood
25 likelihood_pdf = beta.pdf(x, n_W + 1, n_L + 1)
26
27 # make plot
28 plt.figure(figsize=(10, 6))
29 prior_label = f'Prior: Beta({alpha_prior}, {beta_prior})'
30 plt.plot(x, prior_pdf, 'b--', lw=2, label=prior_label)
31 post_label = f'Posterior: Beta({alpha_post}, {beta_post})'
32 plt.plot(x, post_pdf, 'r-', lw=3, label=post_label)
33 likelihood_label = f'Likelihood (scaled): $\propto \theta^{n_W}(1-\theta)^{n_L}$'
34 plt.plot(x, likelihood_pdf, 'g:', lw=2, label=likelihood_label)
35
36 # add vertical lines for means
37 prior_mu_label = f'Prior Mean: {0.6:.2f}'
38 plt.axvline(alpha_prior / (alpha_prior + beta_prior), color='b',
39             linestyle='--', lw=1, label=prior_mu_label)
40 post_mu_label = f'Posterior Mean: {0.55:.2f}'
41 plt.axvline(alpha_post / (alpha_post + beta_post), color='r',
42             linestyle='-', lw=1, label=post_mu_label)
43 like_mle_label = f'MLE (Likelihood Peak): {0.5:.2f}'
44 plt.axvline(n_W / (n_W + n_L), color='g',
45             linestyle=':', lw=1, label=like_mle_label)
46
47 # set args for plot and save
48 plt.xlabel('$\theta$ (Probability of Win)', fontsize=12)
49 plt.ylabel('Density', fontsize=12)
50 plt.title('Bayesian Update: Prior, Likelihood, and Posterior', fontsize=14)
51 plt.legend(fontsize=11)
52 plt.grid(True, linestyle='--', alpha=0.6)
53 plt.xlim(0, 1)
54 plt.savefig('q8_d.png', dpi=150, bbox_inches='tight')

```

Solution 8

Solution 8 (d)

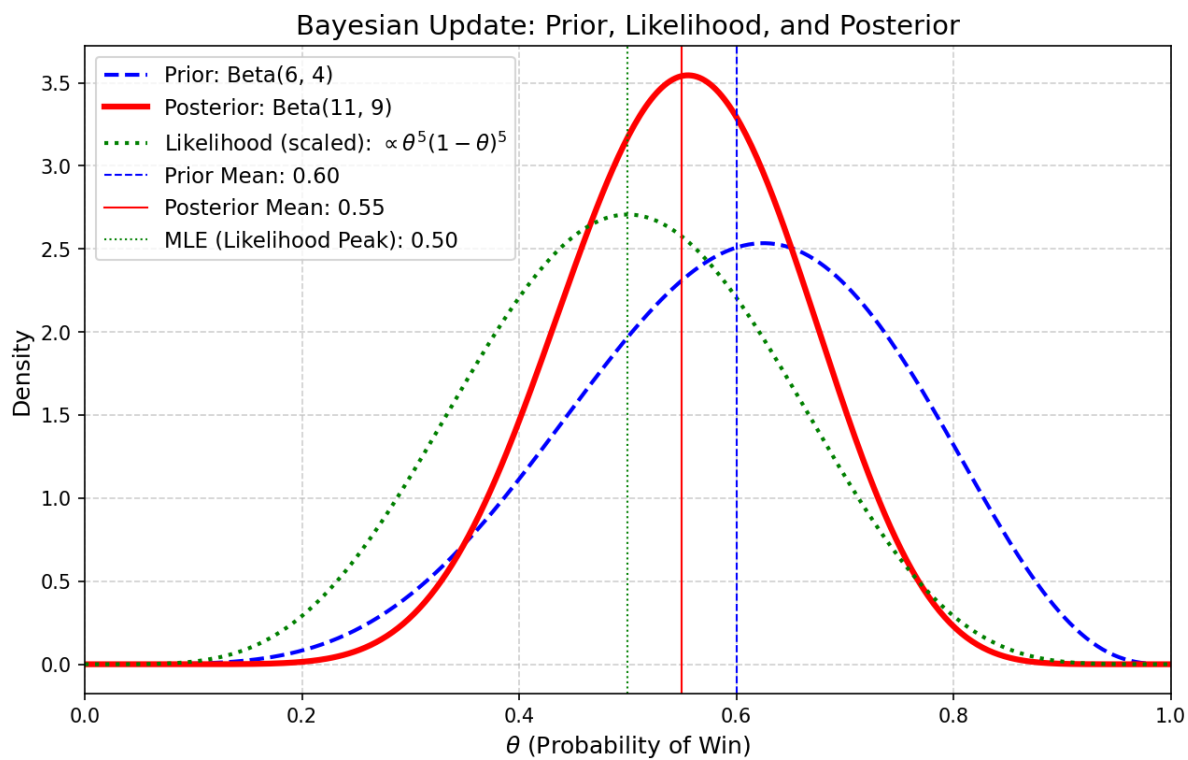


Figure 4: Density plot of the Beta(6, 4) prior (blue, dashed), the Beta(11, 9) posterior (red, solid), and the (scaled) likelihood (green, dotted).

Solution 9

Solution 9 (a)

Let the prior distribution $\theta \sim \text{Gamma}(\alpha, \beta)$, such that $\alpha = 1$ and rate $\beta = 1$.

The expected value (mean) of a Gamma distribution with shape α and rate β is given by the formula:

$$\mathbb{E}[\theta] = \frac{\alpha}{\beta}$$

Substituting our prior parameters:

$$\begin{aligned}\mathbb{E}[\theta] &= \frac{\alpha}{\beta} \\ &= \frac{1}{1} \\ &= 1\end{aligned}$$

$\therefore$ The expected value of θ under the prior is:

$$\mathbb{E}[\theta] = 1$$

Solution 9

Solution 9 (b)

The updated α' and β' are defined as:

$$\alpha' = \alpha + \sum_{i=1}^n x_i \quad \text{and} \quad \beta' = \beta + n$$

Let,

$$\alpha = 1 \quad \beta = 1 \quad n = 10 \quad x = \{13, 2, 4, 20, 10, 9, 4, 5, 12, 15\}$$

Now calculate α' ,

$$\begin{aligned} \alpha' &= \alpha + \sum_{i=1}^n x_i \\ &= 1 + 13 + 2 + 4 + 20 + 10 + 9 + 4 + 5 + 12 + 15 \\ &= 95 \end{aligned}$$

Next calculate β' ,

$$\begin{aligned} \beta' &= \beta + n \\ &= 1 + 10 \\ &= 11 \end{aligned}$$

$\therefore$ The posterior distribution of θ is Gamma(95, 11).

Solution 9

Solution 9 (c)

Prior and Posterior Density Plot Code

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import gamma
4
5 # set prior parameters
6 alpha_prior = 1
7 beta_prior = 1
8
9 # set posterior parameters
10 alpha_post = 95
11 beta_post = 11
12
13 # note: scipy.stats.gamma uses scale = 1/rate
14 # create np.array of x values (theta)
15 x = np.linspace(0, 15, 500)
16
17 # calc the PDF for prior and posterior
18 prior_pdf = gamma.pdf(x, a=alpha_prior, scale=1/beta_prior)
19 post_pdf = gamma.pdf(x, a=alpha_post, scale=1/beta_post)
20
21 # Create the plot
22 plt.figure(figsize=(10, 6))
23 plt.plot(x, prior_pdf, 'b--', lw=2,
24         label=f'Prior: Gamma({alpha_prior}, {beta_prior})')
25 plt.plot(x, post_pdf, 'r-', lw=3,
26         label=f'Posterior: Gamma({alpha_post}, {beta_post})')
27
28 # add vertical lines for the means
29 prior_mean = alpha_prior / beta_prior
30 post_mean = alpha_post / beta_post
31 plt.axvline(prior_mean, color='b', linestyle='--', lw=1,
32             label=f'Prior Mean: {prior_mean:.2f}')
33 plt.axvline(post_mean, color='r', linestyle='-', lw=1,
34             label=f'Posterior Mean: {post_mean:.2f}')
35
36 # set args for plot and save
37 plt.xlabel('$\\theta$ (Mean Accidents per Day)', fontsize=12)
38 plt.ylabel('Density', fontsize=12)
39 plt.title('Bayesian Update: Prior and Posterior Densities for Accident Rate',
40         fontsize=14)
41 plt.legend(fontsize=11)
42 plt.grid(True, linestyle='--', alpha=0.6)
43 plt.xlim(0, 15)
44 plt.ylim(bottom=0)
45 plt.savefig('q9_c_plot.png', dpi=150, bbox_inches='tight')
```

Solution 9

Solution 9 (c)

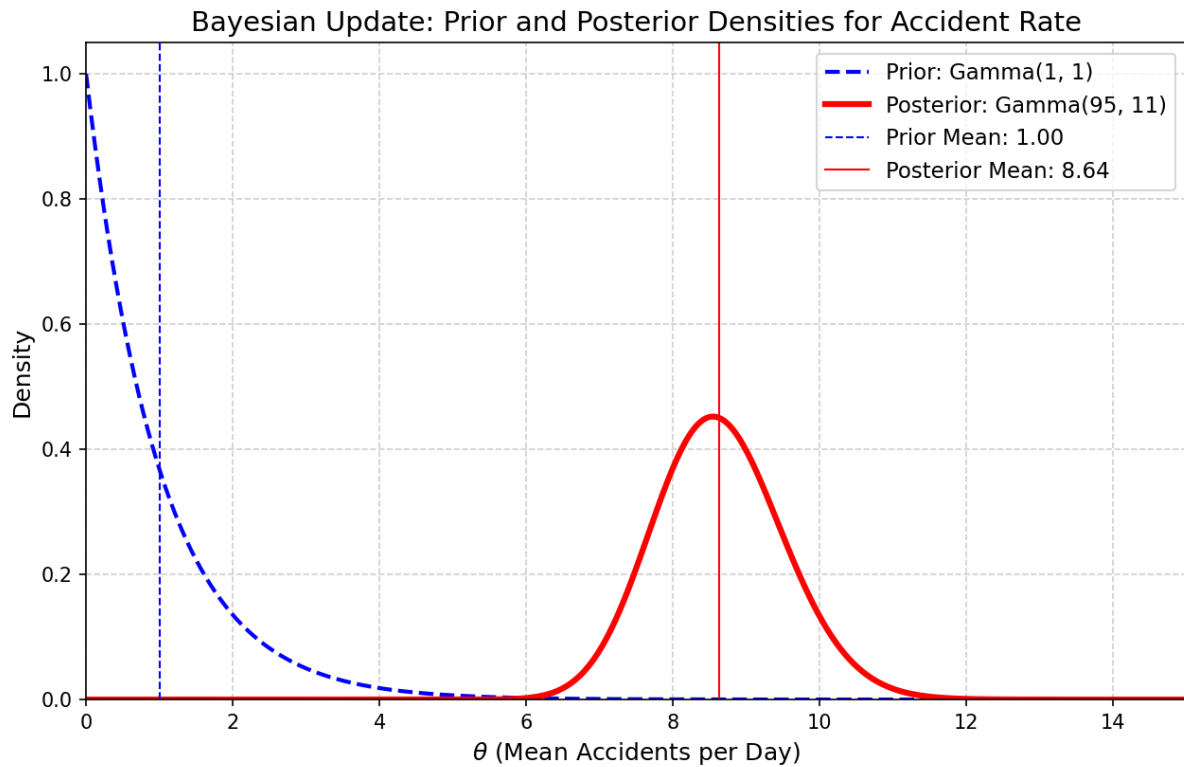


Figure 5: Density plot of the Gamma(1,1) prior (blue, dashed) and the Gamma(95,11) posterior (red, solid). The data has shifted our belief from a prior mean of 1.0 to a posterior mean of ≈ 8.64 , and significantly reduced our uncertainty.

Solution 10

Solution 10 (a)

Let,

$$X = \{A, B, B, A\} \rightarrow n_A = 2, n_B = 2, n_C = 0 \text{ and } n = 4$$

The conjugate prior for a Multinomial likelihood is the Dirichlet distribution:.

$$P(\theta_A, \theta_B, \theta_C) \sim \text{Dirichlet}(\alpha_A, \alpha_B, \alpha_C)$$

A "uniform prior" over the simplex (where $\sum \theta_i = 1$) is a special case of the Dirichlet distribution where all concentration parameters are 1 or:

$$P(\theta_A, \theta_B, \theta_C) \sim \text{Dirichlet}(\alpha_A, \alpha_B, \alpha_C) = \text{Dirichlet}(1, 1, 1)$$

The posterior distribution is also a Dirichlet distribution, to find the posterior we add the observed counts to the prior parameters.

$$\begin{aligned} P((\theta_A, \theta_B, \theta_C) \mid X) &\sim \text{Dirichlet}(\alpha_A + n_A, \alpha_B + n_B, \alpha_C + n_C) \\ &\sim \text{Dirichlet}(1 + 2, 1 + 2, 1 + 0) \\ &\sim \text{Dirichlet}(3, 3, 1) \end{aligned}$$

$\therefore$ the posterior distribution is $\text{Dirichlet}(3, 3, 1)$.

Solution 10

Solution 10 (b)

The formula for the expected value of a component θ_i from a Dirichlet distribution is:

$$\mathbb{E}[\theta_i \mid X] = \frac{\alpha'_i}{\sum_j \alpha'_j}$$

From *part (a)*,

$$\alpha'_A = 3 \quad \alpha'_B = 3 \quad \alpha'_C = 1$$

Now, find the expected value for each candidate's vote share:

$$\begin{aligned}\mathbb{E}[\theta_A \mid X] &= \frac{\alpha'_A}{\alpha'_A + \alpha'_B + \alpha'_C} = \frac{3}{3 + 3 + 1} = \frac{3}{7} \\ \mathbb{E}[\theta_B \mid X] &= \frac{\alpha'_B}{\alpha'_A + \alpha'_B + \alpha'_C} = \frac{3}{3 + 3 + 1} = \frac{3}{7} \\ \mathbb{E}[\theta_C \mid X] &= \frac{\alpha'_C}{\alpha'_A + \alpha'_B + \alpha'_C} = \frac{1}{3 + 3 + 1} = \frac{1}{7}\end{aligned}$$

$\therefore$ The expected value is $\mathbb{E}[(\theta_A, \theta_B, \theta_C) \mid X] = (\frac{3}{7}, \frac{3}{7}, \frac{1}{7})$.
